

End-to-End Sales Forecasting & Demand Intelligence System

Executive Business Report

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Executive Summary

This project analyzes four years of retail sales data to help the business improve inventory planning and demand forecasting. Using historical sales records, we identified sales trends, seasonal patterns, unusual sales events, and product demand behavior. Three forecasting models—SARIMA, Facebook Prophet, and XGBoost—were developed and compared. Among them, the XGBoost model produced the most accurate forecasts with the lowest prediction error. The project also detected unusual sales spikes using anomaly detection and grouped products into demand segments using clustering. These insights can help the supply chain team maintain optimal inventory levels, reduce stock shortages, minimize excess inventory costs, and improve overall business planning.

Key Findings from EDA and Forecasting

Product Performance

The Technology category generated the highest total sales revenue (approximately ₹827,456), followed by Furniture (₹728,659) and Office Supplies (₹705,422). This indicates that Technology products contribute the largest share of business revenue and should receive greater inventory attention.

Regional Performance

Sales increased steadily across all regions during the four-year period. The West region showed the strongest growth trend, while the East region also demonstrated consistent performance. These regions are expected to remain important contributors to future sales.

Seasonality

Monthly sales analysis revealed recurring seasonal patterns. Sales generally increased during the final months of the year, suggesting higher customer demand during festive and holiday shopping periods.

Forecast Model Comparison

Three forecasting models were evaluated using standard error metrics.

Model	MAE	RMSE	MAPE
SARIMA	11,532.90	13,843.49	17.65%
Facebook Prophet	10,128.56	14,561.39	14.33%
XGBoost	8,017.25	13,156.85	10.60%

The XGBoost model achieved the lowest prediction error and was selected as the recommended forecasting model for deployment.

Three-Month Sales Forecast

Based on the XGBoost forecasting model, overall sales are expected to remain stable with gradual growth over the next three months. The model predicts continued demand in the Technology category and stronger sales in the West and East regions. While exact sales values may vary depending on promotions and market conditions, the confidence range suggests that the business should maintain sufficient inventory levels without significant overstocking. Inventory planning should focus particularly on high-demand products to reduce stock-out risks during peak periods.

Top Three Anomalies and Their Likely Causes

The Isolation Forest algorithm detected **11 unusual sales weeks** across the dataset. The three most significant anomalies are likely explained by normal business events rather than system errors.

1. High Sales Spike (Late 2018)

A significant increase in weekly sales was observed during the final months of 2018. This is likely due to festive season shopping, promotional campaigns, or year-end discounts that increased customer purchases.

2. Low Sales Period (Early 2015)

Several weeks recorded unusually low sales during the beginning of 2015. Possible reasons include post-holiday demand reduction, fewer customer orders, or temporary supply limitations.

3. Sudden Mid-Year Sales Increase

An unexpected sales spike occurred during the middle of the dataset. This may have resulted from special promotional offers, bulk corporate purchases, or new product launches.

Product Demand Segmentation and Stocking Strategy

K-Means clustering grouped product sub-categories into four demand segments based on sales volume, growth rate, sales volatility, and average order value.

Cluster 1 – High Volume, Stable Demand

Products in this group consistently generate high sales with relatively stable demand. These products should always remain well stocked because they represent the business's core revenue.

Cluster 2 – Low Volume, Stable Demand

These products experience lower sales but predictable demand. Moderate inventory levels are recommended to avoid unnecessary storage costs while ensuring product availability.

Cluster 3 – Growing Demand

Products in this cluster show increasing customer demand over time. Inventory levels should gradually increase, and marketing efforts may further accelerate sales growth.

Cluster 4 – High Volatility Products

These products experience large fluctuations in demand. Inventory should be monitored closely, and replenishment decisions should rely on frequent sales reviews to avoid excess stock.

Business Recommendations

Recommendation 1 – Increase Inventory for Technology Products

Technology products generated the highest total sales revenue (approximately ₹827,456). Maintaining higher inventory levels for these products can reduce lost sales caused by stock shortages.

Recommendation 2 – Improve Inventory Planning Before Peak Seasons

Historical sales patterns show consistent sales increases during year-end months. Preparing inventory before these seasonal demand peaks can improve customer satisfaction while preventing emergency restocking costs.

Recommendation 3 – Use XGBoost for Demand Forecasting

Among all forecasting models, XGBoost achieved the lowest forecasting error (MAE: 8,017.25 and MAPE: 10.60%). Adopting this model for routine forecasting will improve inventory planning and reduce forecasting inaccuracies.

Risk / Limitation

This forecasting system relies entirely on historical sales data. Unexpected external events such as economic changes, new competitors, supply chain disruptions, pricing changes, or sudden shifts in customer preferences cannot be fully predicted by the model. Therefore, business managers should combine forecasting results with market knowledge and operational planning before making major inventory decisions.

Conclusion

This project successfully developed an end-to-end sales forecasting and demand analysis system using exploratory data analysis, time series forecasting, anomaly detection, clustering, and an interactive Streamlit dashboard. The XGBoost model demonstrated the highest forecasting accuracy and is recommended for production use. By applying these insights, the company can improve inventory management, reduce operational costs, and make more informed business decisions.